

PAUL MITCHELL

SOFTWARE ENGINEERING | CLOUD SYSTEMS & DISTRIBUTED PLATFORMS

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PROFESSIONAL SUMMARY

William & Mary Computer Science M.S. candidate (3.95 GPA, Systems & AI/ML concentration) and B.S. graduate with production-grade Linux systems administration (100+ servers), event-driven backend engineering, and AWS cloud pipelines. Experienced building high-throughput microservices, sub-millisecond API middleware (0.038ms P99), and resilient telemetry architectures with test-driven deployment. Adept in stream processing, cloud security boundaries, and high-reliability software infrastructure.

EDUCATION

College of William & Mary

Williamsburg, VA

Master of Science in Computer Science (AI/ML & Systems Concentration) | Expected Dec 2026 | GPA: 3.95

Bachelor of Science in Computer Science | Graduated Dec 2024

Relevant Coursework: Distributed Systems, Advanced Software Engineering, Machine Learning Systems, Deep Learning, Data Mining

TECHNICAL SKILLS

Cloud & Distributed Systems: AWS (EC2, S3, SQS, Lambda, IAM), Docker, Apache Kafka, Redis, microservices, Linux (Ubuntu/RHEL), Bash

Programming & Back-End: Python, FastAPI, Java, Go, C/C++, SQL (PostgreSQL), Bash, Git/GitLab CI, RESTful APIs, TDD, OOP

Observability & Reliability: Prometheus, Grafana, OpenTelemetry, metrics instrumentation, alert rule engineering, incident triage

Data & Machine Learning: PyTorch, scikit-learn, LightGBM, DistilBERT, Pandas, NumPy, stream processing, anomaly detection

TECHNICAL & PROFESSIONAL EXPERIENCE

IT Administrator | William & Mary Technology Support Center | Williamsburg, VA

Feb 2025 - Present

- Administer and maintain 100+ Linux instructional compute servers, orchestrating automated system configurations via Puppet, DNS management, and network troubleshooting across multi-room labs.
- Architected Prometheus and Grafana telemetry pipelines monitoring cluster uptime, storage utilization, and compute health, cutting incident detection and response times by over 40%.
- Evaluated multi-user Docker containerization architectures, producing technical documentation on isolation boundaries, resource limits, and operational maintainability for department heads.
- Standardized network routing and static IP configurations, maintaining strict configuration control across all production instructional hosts and eliminating configuration drift.
- Developed automated Bash recovery scripts for stalled network daemons, eliminating recurring manual administrator interventions.

Data Analytics & Engineering Extern | Extern, Amazon-Sponsored Project | Remote

Jun 2025 - Sep 2025

- Engineered reproducible automated Python and AWS S3 data pipelines to clean, validate, and structure ~20,000 text records, optimizing data prep execution cycles from hours to minutes.
- Developed a predictive classification model using LightGBM and deployed an interactive Streamlit analytics platform delivering real-time decision-support metrics for program mentors.
- Implemented cloud dataset versioning using DVC on AWS S3 to ensure reproducibility and audit compliance across multi-team evaluations.
- Fine-tuned DistilBERT transformer models ($F1 = 0.89$) to categorize multi-topic survey corpora into prioritized operational action items.

KEY TECHNICAL & SYSTEMS PROJECTS

Distributed Financial Transaction Stream Engine | Python, Kafka, Redis

2026

- Architected a distributed financial stream engine simulating 100k tx/min under chaos-injected 200ms latency spikes and worker failures; guaranteed exactly \$0.00 balance discrepancy using atomic Redis idempotency.
- Engineered adaptive consumer-lag backpressure preventing Kafka partition rebalance storms during network splits; held P99 latency strictly under 6ms with Prometheus/Grafana real-time telemetry.

Real-Time Adversarial Prompt Defense Proxy | Python, FastAPI, Docker

2026

- Engineered an asynchronous reverse proxy evaluating incoming HTTP streams in memory at 0.038ms P99 latency, neutralizing steganographic injection and payload assembly attacks prior to model ingestion.
- Implemented zero-copy token boundary scanners eliminating redundant serialization overhead, achieving 25,000+ req/sec throughput.

Automotive CAN Bus Intrusion Detection & Telemetry System | Python, C

2025

- Processed high-frequency automotive CAN packet streams in real time, transforming raw binary message frames into structured telemetry features for attack classification.
- Validated packet timing anomaly detection algorithms, detecting spoofed frames and bus-flood denial of service attacks with 99.4% precision.